SOLUTIONS: SYSTEMS IDENTIFICATION

1. Solution

a) Based on the block diagram, one gets:

$$\begin{aligned} x_1(t+1) &= 0.5x_1(t) + a_1x_2(t) + x_3(t), \\ x_2(t+1) &= -0.5x_2(t) + 0.1x_4(t) + v_1(t), \\ x_3(t+1) &= a_2x_3(t) + x_4(t) + v_2(t), \\ x_4(t+1) &= a_3x_2(t) + 0.5x_4(t). \end{aligned}$$

Hence, matrices A and B are given by:

$$A = \begin{bmatrix} 0.5 & a_1 & 1 & 0 \\ 0 & -0.5 & 0 & 0.1 \\ 0 & 0 & a_2 & 1 \\ 0 & 0 & a_3 & 0.5 \end{bmatrix}; \quad B = \begin{bmatrix} 0 & 0 \\ 1 & 0 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}$$

This is an easy question. Nevertheless, sometimes students make mistakes on determining matrix B

[3 Marks]

b) Referring to the answer to Question 1-a), it follows that the second and fourth equations evolve independently from the first and the third ones. Hence, we consider these equations separately:

$$\begin{aligned} x_2(t+1) &= -0.5x_2(t) + 0.1x_4(t) + v_1(t), \\ x_4(t+1) &= a_3x_2(t) + 0.5x_4(t). \end{aligned} \tag{1.1}$$

According to (1.1), the stochastic processes $x_2(\cdot)$ and $x_4(\cdot)$ are stationary if all eigenvalues of the matrix

$$M = \begin{bmatrix} -0.5 & 0.1 \\ a_3 & 0.5 \end{bmatrix}$$

lie strictly inside the unit circle. To analyze the eigenvalues of M, we can investigate the roots of $\det(zI - M) = 0$:

$$(z+0.5)(z-0.5) - a_3/10 = 0,$$

or

$$z^2 - 1/4 - a_3/10 = 0.$$

Therefore, the following values for a_3 are acceptable:

$$-1 < \frac{1}{4} + \frac{a_3}{10} < 1$$

which yields

$$-\frac{50}{4} < a_3 < \frac{30}{4}.$$

Answers

Now, since $x_4(\cdot)$ is a stationary process, also $x_4(\cdot)+v_2(\cdot)$ is stationary. Then, we let $\hat{v}_2(\cdot)=x_4(\cdot)+v_2(\cdot)$, and we write

$$x_3(t+1)=a_2x_3(t)+\underbrace{x_4(t)+v_2(t)}_{\hat{v}_2(t)}.$$

Therefore, it can be said that $x_3(\cdot)$ is also a stationary process if

$$-1<a_2<1.$$

Now, since $x_2(\cdot)$ and $x_3(\cdot)$ are stationary process, let us consider the first equation as

$$x_1(t+1)=0.5x_1(t)+\underbrace{a_1x_2(t)+x_3(t)}_{n(t)}$$

where $n(\cdot)$ is a stationary process. Therefore, as $|0.5|<1$, for any finite a_1, one can conclude that $x_1(\cdot)$ is a stationary process as well.

Students often miss the logical connection between stability of the dynamic system that generates the stochastic process and the stationarity of the process itself. Moreover, students sometimes fail to catch that some state equations evolve independently from other state equations which greatly simplifies the analysis.

[6 Marks]

c) According to the answer to Question 1-b), for $a_1=a_2=a_3=0$, $x_1(\cdot)$, $x_2(\cdot)$, $x_3(\cdot)$, and $x_4(\cdot)$ are stationary processes. Therefore, $\mathbb{E}[x_i(t+1)]=\mathbb{E}[x_i(t)], i=1,2,3,4$. Thus, one obtains:

$$\begin{aligned}
m_1&=0.5m_1+m_3,\\
m_2&=-0.5m_2+0.1m_4+\mathbb{E}[v_1(t)],\\
m_3&=m_4+\mathbb{E}[v_2(t)],\\
m_4&=0.5m_4
\end{aligned}$$

which can be simplified as

$$\begin{aligned}
m_1&=0.5m_1+m_3,\\
m_2&=-0.5m_2+0.1m_4,\\
m_3&=m_4+1,\\
m_4&=0.5m_4.
\end{aligned}$$

Therefore, we get $m_1=2$, $m_2=0$, $m_3=1$, and $m_4=0$.

This is a very easy question

[3 Marks]

d) Let us define

Answers

$$
\begin{aligned}
\tilde{x}_1(t) &= x_1(t) - m_1,\\
\tilde{x}_2(t) &= x_2(t) - m_2,\\
\tilde{x}_3(t) &= x_3(t) - m_3,\\
\tilde{x}_4(t) &= x_4(t) - m_4,\\
\tilde{v}_1(t) &= v_1(t),\\
\tilde{v}_2(t) &= v_2(t) - 1.
\end{aligned}
$$

Therefore, from the answer to Question 1-a), for $a_1 = a_2 = a_3 = 0$, we obtain

$$
\begin{aligned}
\tilde{x}_1(t+1) &= 0.5\tilde{x}_1(t) + \tilde{x}_3(t),\\
\tilde{x}_2(t+1) &= -0.5\tilde{x}_2(t) + 0.1\tilde{x}_4(t) + \tilde{v}_1(t),\\
\tilde{x}_3(t+1) &= \tilde{x}_4(t) + \tilde{v}_2(t),\\
\tilde{x}_4(t+1) &= 0.5\tilde{x}_4(t).
\end{aligned}
\tag{1.2}
$$

Now, as the processes $x_1(\cdot)$, $x_2(\cdot)$, $x_3(\cdot)$, and $x_4(\cdot)$ are stationary, we write

$$\mathbb{E}[\tilde{x}_4(t+1)^2] = \mathbb{E}[\tilde{x}_4(t)^2]$$

which from (1.2), gives

$$\mathbb{E}[\tilde{x}_4(t)^2] = 0.25\mathbb{E}[\tilde{x}_4(t)^2],$$

implying that

$$\mathbb{E}[\tilde{x}_4(t)^2] = 0. \tag{1.3}$$

In a similar way, one obtains

$$\mathbb{E}[\tilde{x}_2(t+1)^2] = \mathbb{E}[\tilde{x}_2(t)^2]$$

which from (1.2), gives (note that there is no correlation between $\tilde{x}_i(t)$ and $\tilde{v}_i(t)$)

$$\mathbb{E}[\tilde{x}_2(t)^2] = 0.25\mathbb{E}[\tilde{x}_2(t)^2] + 0.01\mathbb{E}[\tilde{x}_4(t)^2] + \mathbb{E}[v_1(t)^2] - 0.1\mathbb{E}[\tilde{x}_2(t)\tilde{x}_4(t)].$$

In turn, from (1.3) and since $\mathbb{E}[v_1(t)^2] = 1$, it follows that

$$0.75\mathbb{E}[\tilde{x}_2(t)^2] = 1 - 0.1\mathbb{E}[\tilde{x}_2(t)\tilde{x}_4(t)]. \tag{1.4}$$

Moreover, from (1.2) and (1.3), one gets

$$\mathbb{E}[\tilde{x}_2(t+1)\tilde{x}_4(t+1)] = \mathbb{E}[\tilde{x}_2(t)\tilde{x}_4(t)] = -0.25\mathbb{E}[\tilde{x}_2(t)\tilde{x}_4(t)].$$

Therefore,

$$\lambda_{24} = \mathbb{E}[\tilde{x}_2(t)\tilde{x}_4(t)] = 0$$

and then from (1.4),

$$\lambda_{22} = \mathbb{E}[\tilde{x}_2(t)^2] = 1/0.75.$$

To obtain λ_{23}, from (1.2) and (1.3), one obtains (note that there is no correlation between $\tilde{v}_1(t)$ and $\tilde{v}_2(t)$)

$$\lambda_{23} = \mathbb{E}[\tilde{x}_2(t+1)\tilde{x}_3(t+1)] = \mathbb{E}[\tilde{x}_2(t)\tilde{x}_3(t)] = -0.5\mathbb{E}[\tilde{x}_2(t)\tilde{x}_4(t)] + 0.1\mathbb{E}[\tilde{x}_4(t)^2] = 0.$$

Very often students do not use their answers to Question 1-c) and thus do not introduce new random variables in which the means have been removed. Using zero-mean variables greatly simplifies the mathematical derivations.

[8 Marks]

Answers

2. Solution

a) From equations

$$\begin{aligned} x_1(t) &= 0.5x_1(t-1) + x_2(t-1) + v(t), \\ x_2(t) &= -0.1x_2(t-1) + 0.1x_1(t-1), \\ y(t) &= x_1(t), \end{aligned} \tag{2.1}$$

one gets immediately

$$x_1(t+1) = 0.5x_1(t) + x_2(t) + v(t+1)$$

which, since $x_2(t) = -0.1x_2(t-1) + 0.1x_1(t-1)$, can be rewritten as

$$x_1(t+1) = 0.5x_1(t) - 0.1x_2(t-1) + 0.1x_1(t-1) + v(t+1).$$

Moreover, as $x_2(t-1) = x_1(t) - 0.5x_1(t-1) - v(t)$, the ARMA model describing $x_1(t)$ can be rewritten as follows:

$$x_1(t+1) = 0.5x_1(t) - 0.1x_1(t) + 0.05x_1(t-1) + 0.1v(t) + 0.1x_1(t-1) + v(t+1)$$

which by considering $y(t) = x_1(t)$, can be simplified as follows:

$$y(t+1) = 0.4y(t) + 0.15y(t-1) + v(t+1) + 0.1v(t),$$

or

$$y(t) = 0.4y(t-1) + 0.15y(t-2) + v(t) + 0.1v(t-1). \tag{2.2}$$

Easy question. Some students get confused on linking a state equation with an ARMA process described by a difference equation

[3 Marks]

b) First it can be said that the process is stationary as the roots of the characteristic equation of (2.2) in the answer to Question 2-a) $z^2 - 0.4z - 0.15$ are $-59/250$ and $159/250$, and hence they lie inside the unit circle. Moreover, as $\mathbb{E}[v(t)] = 0$, it follows that $\mathbb{E}[y(t)] = 0$. Now, we can obtain $\gamma_y(0)$, $\gamma_y(1)$, and $\gamma_y(2)$ as follows (note that there is no correlation between $y(t)$ and $v(t+\tau), \tau > 0$):

$$\begin{aligned} \mathbb{E}[y(t)^2] = &\frac{4}{25}\mathbb{E}[y(t-1)^2] + \frac{9}{400}\mathbb{E}[y(t-2)^2] + \frac{1}{100}\mathbb{E}[v(t-1)^2] \\ &+ \mathbb{E}[v(t)^2] + \frac{3}{25}\mathbb{E}[y(t-1)y(t-2)] + \frac{2}{25}\mathbb{E}[y(t-1)v(t-1)]. \end{aligned}$$

Since the process $y(\cdot)$ is stationary, from (2.2), it follows that

$$\mathbb{E}[y(t)v(t)] = \mathbb{E}[y(t-1)v(t-1)] = 1.$$

Moreover, again by stationarity arguments:

$$\begin{aligned} \gamma_y(0) &= \mathbb{E}[y(t)^2] = \mathbb{E}[y(t-1)^2], \\ \gamma_y(1) &= \mathbb{E}[y(t)y(t-1)] = \mathbb{E}[y(t-1)y(t-2)], \end{aligned}$$

and therefore, it follows that

$$\gamma_y(0) = \frac{4}{25}\gamma_y(0) + \frac{9}{400}\gamma_y(0) + \frac{101}{100} + \frac{3}{25}\gamma_y(1) + \frac{2}{25}$$

Answers

which can be simplified as follows:

$$\frac{327}{400}\gamma_y(0) - \frac{3}{25}\gamma_y(1) = \frac{109}{100}. \tag{2.3}$$

Moreover, from (2.2) in the answer to Question 2-a) one gets

$$\mathbb{E}[y(t)y(t-1)] = \frac{2}{5}\mathbb{E}[y(t-1)^2] + \frac{3}{20}\mathbb{E}[y(t-1)y(t-2)] + \frac{1}{10}\mathbb{E}[y(t-1)v(t-1)]$$

which implies that

$$\gamma_y(1) = \frac{2}{5}\gamma_y(0) + \frac{3}{20}\gamma_y(1) + \frac{1}{10}$$

and hence

$$\frac{2}{5}\gamma_y(0) - \frac{17}{20}\gamma_y(1) = -\frac{1}{10}. \tag{2.4}$$

Therefore, by solution of (2.3) and (2.4), we get

$$\gamma_y(0) = \frac{3627}{2500}, \gamma_y(1) = \frac{2001}{2500}.$$

Finally,

$$\gamma_y(2) = y(t)y(t-2) = \frac{2}{5}\mathbb{E}[y(t-1)y(t-2)] + \frac{3}{20}\mathbb{E}[y(t-2)^2].$$

Since the process is stationary,

$$\gamma_y(0) = \mathbb{E}[y(t)^2] = \mathbb{E}[y(t-2)^2],$$

and therefore, it can be said that

$$\gamma_y(2) = \frac{2}{5}\gamma_y(1) + \frac{3}{20}\gamma_y(0) = \frac{2689}{5000}.$$

The typical mistake is missing the stationarity analysis and the calculation of the means. Often, students make calculation mistakes in solving the linear system of equations

[5 Marks]

c) By considering the ARMA model obtained in the answer to Question 2-a) we write

$$A(z)y(t) = C(z)v(t)$$

where

$$A(z) = 1 - \frac{2}{5}z^{-1} - \frac{3}{20}z^{-2},$$
$$C(z) = 1 + \frac{1}{10}z^{-1}.$$

Therefore,

$$\frac{|C(z)|^2}{|A(z)|^2}\Big|_{z=e^{j\omega}}.1 = \frac{|z^2 + \frac{1}{10}z|^2}{|z^2 - \frac{2}{5}z - \frac{3}{20}|^2}\Big|_{z=e^{j\omega}}$$

$$= \frac{(z^2 + \frac{1}{10}z)(z^{-2} + \frac{1}{10}z^{-1})}{(z^2 - \frac{2}{5}z - \frac{3}{20})(z^{-2} - \frac{2}{5}z^{-1} - \frac{3}{20})}\Big|_{z=e^{j\omega}}$$

Answers

which can be rewritten as

$$\Gamma_y(\omega) = \frac{\frac{101}{100} + \frac{1}{10}z + \frac{1}{10}z^{-1}}{\frac{473}{400} - \frac{17}{50}z^{-1} - \frac{3}{20}z^{-2} - \frac{3}{20}z^{2} - \frac{17}{50}z}\Big|_{z=e^{j\omega}}$$

$$= \frac{\frac{101}{100} + \frac{1}{5}\cos(\omega)}{\frac{473}{400} - \frac{17}{25}\cos(\omega) - \frac{3}{10}\cos(2\omega)}.$$

Even if the question is very easy, several students fail to write the transfer function at the beginning thus missing the possibility of using the simple formula for the spectrum presented during the lectures and described in the handouts with plenty of examples

[3 Marks]

d) By considering $A(z)$ and $C(z)$ obtained in the answer to Question 2-c) we write

$$\hat{y}(t+1|t) = \frac{C(z)-A(z)}{C(z)}y(t+1) = \frac{0.5z^{-1}+0.15z^{-2}}{1+0.1z^{-1}}y(t+1)$$

which can be rewritten as

$$\hat{y}(t+1|t) = \frac{0.5+0.15z^{-1}}{1+0.1z^{-1}}y(t)$$

thus yielding

$$\hat{y}(t+1|t) = -0.1\hat{y}(t|t-1) + 0.5y(t) + 0.15y(t-1).$$

Easy question but quite often students write the predictor not as a function of the observed variable y but in terms of the noise process v

[3 Marks]

e) By carrying out two steps of the division of $C(z)$ by $A(z)$, the first step gives

$$\frac{C(z)}{A(z)} = \frac{1+0.1z^{-1}}{1-0.4z^{-1}-0.15z^{-2}} = 1 + z^{-1}\frac{0.5+0.15z^{-1}}{1-0.4z^{-1}-0.15z^{-2}}$$

and then, in the next step one gets

$$\frac{C(z)}{A(z)} = 1 + 0.5z^{-1} + z^{-2}\frac{0.35+0.075z^{-1}}{1-0.4z^{-1}-0.15z^{-2}}. \tag{2.5}$$

Therefore, since $v(t) = A(z)/C(z)y(t)$, the optimal two-steps ahead predictor $\hat{y}(t+2|t)$ of $y(t)$ can be written as follows:

$$\hat{y}(t+2|t) = \frac{0.35+0.075z^{-1}}{1+0.1z^{-1}}y(t)$$

which yields

$$\hat{y}(t+2|t) = -0.1\hat{y}(t+1|t) + 0.35y(t) + 0.075y(t-1).$$

In addition to the typical mistake highlighted for the previous question, some students also make wrong calculations in two-steps polynomial division

[3 Marks]

Answers

f) From (2.5) in the answer to Question 2-e), it follows that

$$\varepsilon(t) = y(t+2) - \hat{y}(t+2|t) = (1+0.5z^{-1})v(t+2)$$

implying

$$\varepsilon(t) = y(t+2) - \hat{y}(t+2|t) = v(t+2) + 0.5v(t+1).$$

Thus,

$$\text{var}(\varepsilon) = 1 + 0.25 = 1.25.$$

The typical mistake here is not realising that the prediction error is a MA process with a trivial calculation of the variance

[3 Marks]

Answers

3. Solution

a) Based on the discretized form of $\dot{\theta}(t)$ of $\ddot{\theta}(t)$, the discretized form of the model can be immediately obtained as

$$\begin{aligned}\tau(t) =& ml^2(\theta(t+2)-2\theta(t+1)+\theta(t))+c_1(\theta(t+1)-\theta(t))\\ &+c_2\text{sign}(\theta(t))+mlg\cos(\theta(t)).\end{aligned} \tag{3.1}$$

Even if all discretised variables have been clearly identified in the text of the exam paper, students often fail to understand after replacing the given definitions into the original differential equation, obtaining the difference equation is just a matter of easy algebraic manipulations

[4 Marks]

b) Since m, l, c_1, c_2 are unknown, the unknown parameters of (3.1) in the answer to Question 3-a) can be chosen as

$$\begin{aligned}\alpha_1 &= ml^2,\\ \alpha_2 &= c_1,\\ \alpha_3 &= c_2,\\ \alpha_4 &= ml.\end{aligned}$$

Therefore, $N = 4$ and accordingly

$$\begin{aligned}\rho_1(t) &= \theta(t+2)-2\theta(t+1)+\theta(t),\\ \rho_2(t) &= \theta(t+1)-\theta(t),\\ \rho_3(t) &= \text{sign}(\theta(t)),\\ \rho_4(t) &= g\cos(\theta(t)).\end{aligned}$$

Thus, the parametric model of $\tau(t)$ can be stated as follows:

$$\tau(t) = \sum_{i=1}^{4}\alpha_i\rho_i(t).$$

This is a more difficult question. The key is obtaining a linear-in-parameters expression for τ and to do so a suitable definition of the regressors ρ_i is key. Several students had issues on realising how to define the regressors.

[8 Marks]

c) According to the parametric model obtained in the answer to Question 3-b), one considers the estimation of $\tau(t)$ as $\hat{\tau}(t,\hat{\alpha})$ which can be stated as a multiplication of a known nonlinear vector and an unknown parameters vector as follows:

$$\hat{\tau}(t,\hat{\alpha}) = \Psi(t)^\top\hat{\alpha}$$

where the following definitions hold:

$$\Psi(t) = \begin{bmatrix}\theta_m(t+2)-2\theta_m(t+1)+\theta_m(t)\\ \theta_m(t+1)-\theta_m(t)\\ \text{sign}(\theta_m(t))\\ g\cos(\theta_m(t))\end{bmatrix},$$

$$\hat{\alpha} = \begin{bmatrix}\hat{\alpha}_1 & \hat{\alpha}_2 & \hat{\alpha}_3 & \hat{\alpha}_4\end{bmatrix}^\top.$$

Answers

To estimate α, we consider the following estimation error:

$$e(j) = \tau_m(j) - \hat{\tau}(t(j), \hat{\alpha}) = \tau_m(j) - \Psi(t(j))^\top \hat{\alpha}, j = 1, 2, \ldots, M-2,$$

where $t(j)$ denotes the time associated with the jth sample. Now, we consider a cost function of the form:

$$J(\hat{\alpha}) = \frac{1}{M-2} \sum_{j=1}^{M-2} e(j)^2 = \frac{1}{M-2} \sum_{j=1}^{M-2} \big(\tau_m(j) - \Psi(t(j))^\top \hat{\alpha}\big)^2.$$

The objective is to obtain the estimated values $\hat{\alpha} = \begin{bmatrix}\hat{\alpha}_1 & \hat{\alpha}_2 & \hat{\alpha}_3 & \hat{\alpha}_4\end{bmatrix}^\top$ subject to $\min_{\hat{\alpha}} J(\hat{\alpha})$. Therefore, it is sufficient to find $\hat{\alpha}$ making $\frac{\partial J(\hat{\alpha})}{\partial \hat{\alpha}} = [0\ 0\ 0\ 0]$. We have

$$\begin{aligned}\frac{\partial J(\hat{\alpha})}{\partial \hat{\alpha}} &= \frac{2}{M-2} \sum_{j=1}^{M-2} \big(\tau_m(j) - \Psi(t(j))^\top \hat{\alpha}\big) \frac{\partial}{\partial \hat{\alpha}} \big(\tau_m(j) - \Psi(t(j))^\top \hat{\alpha}\big) \\ &= -\frac{2}{M-2} \sum_{j=1}^{M-2} \big(\tau_m(j) - \Psi(t(j))^\top \hat{\alpha}\big) \Psi(t(j))^\top.\end{aligned}$$

Now, to make $\frac{\partial J(\hat{\alpha})}{\partial \hat{\alpha}} = [0\ 0\ 0\ 0]$, we should have

$$\sum_{j=1}^{M-2} \Psi(t(j)) \Psi(t(j))^\top \hat{\alpha} = \sum_{j=1}^{M-2} \tau_m(i) \Psi(t(j)).$$

Therefore, if the matrix $\sum_{j=1}^{M-2} \Psi(t(j)) \Psi(t(j))^\top$ is full rank, $\hat{\alpha}$ can be obtained as follows:

$$\hat{\alpha} = \Big(\sum_{j=1}^{M-2} \Psi(t(j)) \Psi(t(j))^\top \Big)^{-1} \sum_{j=1}^{M-2} \tau_m(i) \Psi(t(j)).$$

Now, the values of $\hat{l} = \hat{\alpha}_1 / \hat{\alpha}_4$, $\hat{c}_1 = \hat{\alpha}_2$, $\hat{c}_2 = \hat{\alpha}_3$, and $\hat{m} = \hat{\alpha}_4 / \hat{l}$ can be obtained.

If the answer to the previous question is correct, answering this question takes some time but does not present any conceptual difficulty, it is just a standard Least-Squares procedure

[8 Marks]

Answers

4. Solution

a) A moving average model with order n has a correlation function $\gamma_v(\tau)$ which is zero for $|\tau| > n$. Therefore, we should analyze the empirical estimate of the correlation function as follows:

$$\hat{\gamma}_v(\tau) = \frac{1}{6-\tau}\sum_{i=0}^{5-\tau} v(i)v(i+\tau)$$

yielding

$$\hat{\gamma}_v(0) = 91.065, \hat{\gamma}_v(1) = 13.9, \hat{\gamma}_v(2) = 9.6125, \hat{\gamma}_v(3) = -7.8333, \hat{\gamma}_v(4) = 4.5, \hat{\gamma}_v(5) = 17.$$

Therefore, basing on the available samples, it can be concluded that an MA model cannot model the process.

Easy question. Nevertheless, some student did not use the hint given in the text of the exam paper that is, they do not proceed with the empirical estimate and instead used the expectation, which is wrong

[4 Marks]

b) The model in prediction form can be stated as

$$\hat{v}(t|t-1) = \hat{a}v(t-1).$$

Now, to estimate a as $\hat{a}$, we consider a cost function for estimation as follows:

$$J(\hat{a}) = \frac{1}{5}\sum_{t=1}^{5}[v(t) - \hat{v}(t|t-1)]^2 = \frac{1}{5}\sum_{t=1}^{5}[v(t) - \hat{a}v(t-1)]^2.$$

Therefore, the optimal value $\hat{a}$ is the argument of the minimum of the cost function which can be obtained as follows:

$$\frac{\partial J(\hat{a})}{\partial \hat{a}} = -\frac{2}{5}\sum_{t=1}^{5}[v(t)v(t-1) - \hat{a}v(t-1)^2] = 0$$

thus getting

$$\hat{a} = \frac{\sum_{t=1}^{5} v(t)v(t-1)}{\sum_{t=1}^{5} v(t-1)^2} = \frac{69.5}{543.5} = 0.1279.$$

This is a standard scalar Least-Squares solution. Several students just made some numerical mistakes in the calculations

[6 Marks]

c) The model in the prediction form can be stated as

$$\hat{v}(t|t-1) = \hat{a}v(t-1) + \hat{b}v(t-2).$$

Now, to estimate a as $\hat{a}$ and b as $\hat{b}$, we consider a cost function for estimation as follows:

$$J(\hat{a},\hat{b}) = \frac{1}{5}\sum_{t=2}^{5}[v(t) - \hat{v}(t|t-1)]^2 = \frac{1}{5}\sum_{t=2}^{5}[v(t) - \hat{a}v(t-1) - \hat{b}v(t-2)]^2.$$

Answers

Therefore, the optimal $(\hat{a},\hat{b})$ is the argument of the minimum of the cost function which can be obtained as follows:

$$\frac{\partial J(\hat{a},\hat{b})}{\partial \hat{a}} = -\frac{2}{5}\sum_{t=2}^{5}[v(t)v(t-1) - \hat{a}v(t-1)^2 - \hat{b}v(t-1)v(t-2)] = 0,$$
$$\frac{\partial J(\hat{a},\hat{b})}{\partial \hat{b}} = -\frac{2}{5}\sum_{t=2}^{5}[v(t)v(t-2) - \hat{a}v(t-1)v(t-2) - \hat{b}v(t-2)^2] = 0.$$

Hence:

$$-130.5 - 443.5\hat{a} - 73.75\hat{b} = 0,$$
$$38.45 - 73.75\hat{a} - 537.25\hat{b} = 0.$$

Therefore, one gets

$$\hat{a} = -0.3133, \hat{b} = 0.1146.$$

Analogously to the previous question, this is a standard Least-Squares solution but in this case two parameters have to be determined. Several students just made some numerical mistakes in the calculations

[6 Marks]

d) Performance indexes can be taken as the cost functions derived in the answers to Questions 4-b) and 4-c) evaluated on the calculated optimal values of the parameters that is: $J(\hat{a})$ for $\hat{a} = 0.1279$ and $J(\hat{a},\hat{b})$ for $(\hat{a},\hat{b}) = (-0.3133, 0.1146)$. Hence,

$$J(0.1279) = 87.5,$$
$$J(-0.3133, 0.1146) = 0.2746.$$

These values of the performance indexes clearly show the improvement using a higher-order model.

Straightforward question once the previous two questions have been answered. Some students just incurred in calculation mistakes

[4 Marks]

Answers